^{36}Ar(pol d,^{3}He) 1993Ma50

$J^{\pi}=0^{+}$ for ^{36}Ar ground state.

1993Ma50: a 52-MeV polarized deuteron beam at 15 nA was produced from the Karlsruhe isochronous cyclotron at the Max-Plank-Institut fur Kernphysik. The target was 350-mbar 99.86% enriched ^{36}Ar gas. Reaction products were detected using a Si detector telescope consisting of a 250−μm ΔE-strip detector and a 1.5 -mm surface-barrier E detector with FWHM=130 keV. Measured angular distributions of cross sections σ(E(^{3}He),θ) and angular distributions of analyzing powers iT_{11}(E(^{3}He),θ). Deduced levels, J, π, L-transfers, and spectroscopic factors from DWBA analysis of σ(E(^{3}He),θ) and iT_{11}(E(^{3}He),θ).

1993Ma50 states that spectroscopic factors were obtained by fitting the forward-angle maxima. The agreement with the analyzing powers is only qualitative. The determination of spins and parities of final states is mainly based on a comparison with measured angular distributions leading to final states with known spins. The comparison with DWBA predictions serves as additional check. The data were taken only in a relatively small angle interval. This is sufficient for the spin determination but not favorable for the measurement of spectroscopic factors.

^{35}Cl Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=2.95 is a normalization factor used by 1993Ma50 from 1966Ba54.

E(level)	J$^{\pi}$[@]	L[@]	C^{2}S[@]	Comments
0	3/2$^+$[†]	2	1.72[†]	
1213 *4*	1/2$^+$	0	0.78	J$^{\pi}$: $2s_{1/2}$.
1785[#] *50*				J$^{\pi}$: (5/2$^+$) is given in Table 2 of 1993Ma50. C^{2}S: 0.04 for $J^{\pi}=5/2^{+}$ (1993Ma50).
2676[#] *7*				J$^{\pi}$: (3/2$^+$) is given in Table 2 of 1993Ma50. C^{2}S: 0.16 for $J^{\pi}=3/2^{+}$ (1993Ma50).
3002 *6*	5/2$^+$[‡]	2	1.32[‡]	
3159 *20*	7/2$^-$	3	0.19	J$^{\pi}$: $1f_{7/2}$.
3948 *40*	(3/2$^+$)[†]	2	0.03[†]	E(level): possible doublet of 3918.48 and 3967.5 (1990En08) suggested by Table 2 of 1993Ma50. J$^{\pi}$: (3/2$^+$) and (1/2$^+$) are given in Table 2 of 1993Ma50. Evaluators adopt (3/2$^+$) based on DWBA fit of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in Fig. 3 of 1993Ma50. C^{2}S: for $J^{\pi}=3/2^{+}$; 0.08 for $J^{\pi}=1/2^{+}$ (1993Ma50).
4839[#] *12*				E(level): possible doublet of 4839.09 (1/2$^+$,3/2) (1990En08) and an unspecified (1/2$^+$) component, suggested by Table 2 of 1993Ma50. J$^{\pi}$: (5/2$^+$) and (1/2$^+$) are given in Table 2 of 1993Ma50. C^{2}S: 0.08 for $J^{\pi}=5/2^{+}$; 0.03 for $J^{\pi}=1/2^{+}$ (1993Ma50).
5155 *11*	(5/2$^+$)[‡]	2	0.12[‡]	J$^{\pi}$: (7/2$^-$) and 5/2$^+$ are given in Table 2 of 1993Ma50. Evaluators adopt (5/2$^+$), based on DWBA fit of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in Fig. 3 of 1993Ma50.
5583 *13*	5/2$^+$[‡]	2	0.38[‡]	
5720 *11*	5/2$^+$[‡]	2	1.10[‡]	
6136 *8*	5/2$^+$[‡]	2	0.63[‡]	
6745 *12*	(1/2$^+$,5/2$^+$)	(0,2)		J$^{\pi}$: $2s_{1/2}$ is given in Fig. 4; 5/2$^+$ and 1/2$^+$ are given in Table 2 of 1993Ma50. C^{2}S: 0.28 for $J^{\pi}=5/2^{+}$; 0.09 for $J^{\pi}=1/2^{+}$ (1993Ma50).
6953 *43*	5/2$^+$[‡]	2	0.49[‡]	
7181 *60*	5/2$^+$[‡]	2	0.20[‡]	
7565 *20*	5/2$^+$[‡]	2	0.30[‡]	
8007 *36*	(1/2$^-$,5/2$^+$)	(1,2)		J$^{\pi}$: $1p_{1/2}$ is given in Fig. 4; 5/2$^+$ and 1/2$^-$ are given in Table 2 of 1993Ma50. C^{2}S: 0.20 for $J^{\pi}=5/2^{+}$; 0.16 for $J^{\pi}=1/2^{-}$ (1993Ma50).
8204 *34*	5/2$^+$[‡]	2	0.14[‡]	
8580[#] *20*				J$^{\pi}$: 5/2$^+$ is given in Table 2 of 1993Ma50. C^{2}S: 0.22 for $J^{\pi}=5/2^{+}$ (1993Ma50).
8921[#] *70*				J$^{\pi}$: 5/2$^+$ is given in Table 2 of 1993Ma50.

Continued on next page (footnotes at end of table)

^{36}Ar(pol d,^{3}He) 1993Ma50 (continued)

^{35}Cl Levels (continued)

E(level)	J^{π} [@]	L [@]	Comments
			C^2S: 0.18 for $J^{\pi}=5/2^+$ (1993Ma50).
9.22×10^3 [#] *12*			J^{π}: $5/2^+$ is given in Table 2 of 1993Ma50. C^2S: 0.12 for $J^{\pi}=5/2^+$ (1993Ma50).
9514 *20*	$(1/2^-,5/2^+)$	(1,2)	J^{π}: $1p_{1/2}$ is given in Fig. 4; $5/2^+$ and $1/2^-$ are given in Table 2 of 1993Ma50. C^2S: 0.30 for $J^{\pi}=5/2^+$; 0.18 for $J^{\pi}=1/2^-$ (1993Ma50).
9.82×10^3 *12*	$(1/2^-,5/2^+)$	(1,2)	J^{π}: $1p_{1/2}$ is given in Fig. 4; $5/2^+$ and $1/2^-$ are given in Table 2 of 1993Ma50. C^2S: 0.26 for $J^{\pi}=5/2^+$; 0.10 for $J^{\pi}=1/2^-$ (1993Ma50).
1.25×10^4 *5*			$1d_{5/2}$ is given for a range of 12.0-13.0 MeV in Fig. 4 of 1993Ma50.

† L-1/2 transfer from analyzing power measurements; $1d_{3/2}$ proton transfer assumed in DWBA calculations.

‡ L+1/2 transfer from analyzing power measurements; $1d_{5/2}$ proton transfer assumed in DWBA calculations.

No data on $\sigma(E(^3He),\theta)$, $iT_{11}(E(^3He),\theta)$, or L-transfer is given.

@ From DWBA analysis of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in 1993Ma50.